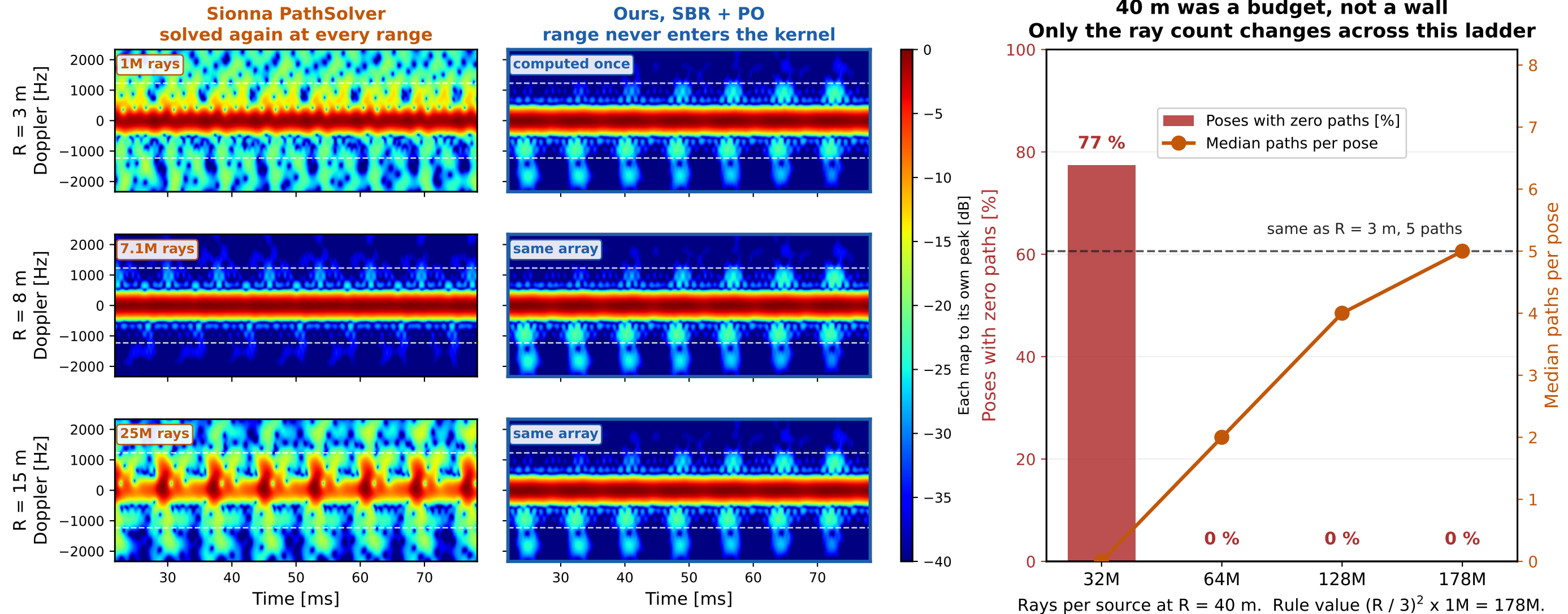


DJI Matrice 4E, belly view, 3.5 GHz. Left, one row per range, the same 60 ms window. Right, what it costs Sionna to reach 40 m.



Left. Only the Sionna column is solved again at each range, with the ray budget raised as the target shrinks in solid angle. Our column is computed once on a target anchored plane wave grid, so the same array is drawn in all three rows. That repetition is the claim, not an accident of plotting. Far field boundary 2D squared over lambda is about 8.0 m, so the 3 m row is near field for Sionna while 8 and 15 m are far field. No pose is ray starved at 3, 8 or 15 m. Maps are normalized to their own peak, so this compares structure only, never level.

Right. At 40 m a starved budget leaves most poses with no path at all, which looks like a collapse. Raising the ray count alone removes it, and at the rule value the path count matches the 3 m case. A separate run at the same 32M budget gave 78 % empty poses, close to the 32M bar here. Ledger outputs/report07_ray_budget_test.json, 128 poses per rung, 5 kHz.